Lecture 11: Linear ODE with variable coefficients (Cauchy-Euler Equation)

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = g(x)$$

First we will make them constant coefficients by letting $x = e^t$ $\therefore t = \ln x$ $\therefore \frac{dt}{dx} = \frac{1}{x}$

$$\therefore y' = \frac{dy}{dx} = \frac{dy}{dt}\frac{dt}{dx} = \frac{dy}{dt}\frac{1}{x} = \boxed{\frac{1}{x}\frac{dy}{dt}} = \frac{1}{x} Dy \quad \text{--- (*)}$$

$$\boxed{D = \frac{d}{dt}}$$

$$\boxed{xy' =} \boxed{x\frac{dy}{dx} = Dy}$$

$$y'' = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dt}\right) = \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dt}\right) + \frac{dy}{dt}\left(-\frac{1}{x^2}\right)$$

$$\therefore \frac{d^2y}{dx^2} = \frac{1}{x}\frac{d}{dt}\left(\frac{dy}{dt}\right)\frac{dt}{dx} - \frac{1}{x^2}\frac{dy}{dt}$$

$$= \frac{1}{x}\frac{d^2y}{dt^2}\cdot\frac{1}{x} - \frac{1}{x^2}\frac{dy}{dt}$$

$$= \frac{1}{x^2}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)$$

$$\therefore x^2\frac{dy}{dt} = \frac{d^2y}{dt^2} - \frac{dy}{dt} = D^2y - Dy$$

$$\boxed{x^2y'' =} \boxed{x^2\frac{dy}{dt} = D(D-1)y}$$

Similarly, $x^3\frac{d^3y}{dx^3} = D(D-1)(D-2)y$ $\boxed{D = \frac{d}{dt}}$

$$\Rightarrow x^3y''' = D(D-1)(D-2)y$$

Example 01: Solve $x^2y'' - 3xy' + 3y = 2x^4e^x$

Ans: Given $x^2y'' - 3xy' + 3y = 2x^4e^x$ --- (1)

Let $x = e^t \Rightarrow t = \ln x$

$$xy' = Dy$$

$$x^2y'' = D(D-1)y \quad \text{Here } D = \frac{d}{dt}$$

$$(1) \Rightarrow D(D-1)y - 3Dy + 3y = 2(e^t)^4 \cdot e^{et}$$

$$\Rightarrow [D^2 - D - 3D + 3]y = 2e^{4t}e^{et}$$

$$\Rightarrow (D^2 - 4D + 3)y = 2e^{4t}e^{et} \quad \text{--- (2)}$$

A.E. $m^2 - 4m + 3 = 0$

$\therefore m = 1, 3$

$$y_c = c_1e^t + c_2e^{3t} = c_1x + c_2x^3$$

Here $y_1 = x$, $y_2 = x^3$

Note: R.H.S of (2) is not suitable for inverse operator method due to (e^{et}) term, but it also is difficult for VPM (variation of parameters method)

Now we will use (1) for getting y_p.

$$(1) \Rightarrow y'' - \frac{3}{x}y' + \frac{3}{x^2}y = \underline{2xe^x} = f(x) \quad \text{--- (3)}$$

$$W = \begin{vmatrix} x & x^3 \\ 1 & 3x^2 \end{vmatrix} = 3x^3 - x^3 = 2x^3 \neq 0$$

$$u_1 = -\int \frac{y_2 f(x)}{W}\,dx = -\int \frac{\cancel{x^3} \cdot \cancel{2x^2} e^x}{\cancel{2x^3}}\,dx = -\int \frac{x^2 e^x}{1}\,dx$$

$$u_1 = -(x^2 e^x - 2xe^x + 2e^x)$$

Rep. Diff.		Rep. Int.
x^2	$+$	e^x
$2x$	$-$	e^x
2	$+$	e^x
0		e^x

$= x^2 e^x - 2xe^x + 2e^x$

$$u_2 = \int \frac{y_1 f(x)}{W}\,dx$$

$$= \int \frac{x \cancel{2x^2} e^x}{\cancel{2x^3}}\,dx$$

$$= \int e^x\,dx = e^x$$

$$\therefore\ y_p = u_1 y_1 + u_2 y_2 = -x(x^2 e^x - 2xe^x + 2e^x) + x^3 e^x$$

$$= \cancel{-x^3 e^x} + 2x^2 e^x - 2xe^x + \cancel{x^3 e^x}$$

$$= 2x^2 e^x - 2xe^x$$

G.S. $y_p = y_c + y_p = c_1 x + c_2 x^3 + 2x^2 e^x - 2xe^x$ Ans

Example 02: Solve $x^2 y'' - xy' + y = \ln x$

Ans: Given $x^2 y'' - xy' + y = \ln x$ ---- (1)

Let $x = e^t \Rightarrow t = \ln x$

$$xy' = Dy$$

$$xy'' = D(D-1)y \quad \text{Here } D = \frac{d}{dt}$$

(1) $\Rightarrow$ $D(D-1)y - Dy + y = t$

$\Rightarrow (D^2 - 2D + 1)y = t$ ---- (2)

A.E. $m^2-2m+1=0$

$\Rightarrow (m-1)^2=0$

$m=1,1$

$y_c = c_1 e^t + c_2 t e^t = c_1 x + c_2 x \ln x$

<u>Note</u>: R.H.S of (2) is a polynomial function (monomial) of t, so we can apply rule I of inverse operator method for getting y_p.

(2) $\Rightarrow$ $y_p = \frac{1}{D^2-2D+1} t = \frac{1}{(D-1)^2} t = \frac{1}{(1-D)^2} t$

$y_p = (1-D)^{-2} t = (1+2D+\cdots) t$

$= t+2 = 2+\ln x$

G.S. $y = y_c + y_p = c_1 x + c_2 x \ln x + 2 + \ln x$

Ans

<u>Example 03</u>: Solve $\{(x+2)^2 D^2 - (x+2)D+1\} y = 3x+4$

<u>Ans</u>: Given $[(x+2)^2 D^2 - (x+2)D+1] y = 3x+4$ --- (1)

Let $(x+2) = e^t \Rightarrow x = e^t - 2 \Rightarrow \frac{dx}{dt} = e^t$

$\frac{dy}{dx} = \frac{dy}{dt}\frac{dt}{dx} = \frac{dy}{dt}\frac{1}{e^t} = \boxed{e^{-t}\frac{dy}{dt}}$

$\Rightarrow \boxed{\frac{dy}{dx} = e^{-t}\bar{D}y}$ $\boxed{\bar{D} = \frac{d}{dt}}$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left(e^{-t}\frac{dy}{dt}\right)\frac{dt}{dx}$$

$$= e^{-t}\left(e^{-t}\frac{d^2y}{dt^2} - e^{-t}\frac{dy}{dt}\right)$$

$$= e^{-2t}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)$$

$$= e^{-2t}\left(D^2y - Dy\right)$$

$$= e^{-2t}D(D-1)y$$

(1) $\Rightarrow$ $e^{2t}e^{-2t}D(D-1)y - e^{t}e^{-t}Dy + y = 3(e^{t}-2)+4$

$\Rightarrow$ $D(D-1)y - Dy + y = 3e^{t} - 2$

$\Rightarrow$ $(D^2 - 2D + 1)y = 3e^{t} - 2$ —— (2)

A.E, $m^2 - 2m + 1 = 0$ $m = 1, 1$

$y_c = c_1e^{t} + c_2te^{t} = \cancel{c_1\ln(x+2)+c_2}$

$= c_1(x+2) + c_2(x+2)\ln(x+2)$ $\left[\begin{array}{l} x+2 = e^{t} \\ t = \ln(x+2) \end{array}\right]$

(2) $\Rightarrow$ $y_p = \frac{1}{D^2-2D+1}3e^{t} - \frac{1}{D^2-2D+1}2$

$= 3e^{t}\frac{1}{(D+1)^2-2(D+1)+1}1 - 2\frac{1}{[1+(D^2-2D)]}1$

$= 3e^{t}\frac{1}{D^2+\cancel{2D}+\cancel{1}-\cancel{2D}-\cancel{2}+\cancel{1}}1 - 2\left(1-(D^2+2D)+\cdots\right]1$

$$y_p = 3e^{t} \frac{1}{D^2} 1 - 2 \cdot 1$$

$$= e^{3t} \frac{1}{D}(t) - 2$$

$$= e^{3t} \frac{t^2}{2} - 2$$

$$= \frac{t^2 3e^{t}}{2} - 2$$

Alternative rule 2 for failure case

$$y_p = \frac{1}{D^2 - 2D + 1} 3e^{t}$$

$$= 3t \frac{1}{2D-2} e^{t}$$

$$= 3t^2 \frac{1}{2} e^{t}$$

G.S. $y = c_1(x+2) + c_2(x+2)\ln(x+2) + \frac{3t^2}{2} e^{t} - 2$

Ans

Home task: MKC book: P-117

Example: 4.19, 4.20, 4.21, 4.24

Exercise: P-122 20, 22, 23, 26

Zills book: Exercise 4.7 P-168

19, 21, 22, 23, 24, 34, 36.